Statement of Purpose: Amazon ML Summer School 2026

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Current Focus: AI-Powered Civic Governance & Intelligence

My technical journey in AI/ML is defined by the transition from understanding algorithms to architecting systems that solve systemic challenges. As a Computer Science student at Terna Engineering College, I have moved beyond simple model implementation to building Nivaran AI, a sophisticated internal intelligence platform designed for government administrators to manage civic grievances at scale.

In building Nivaran, I developed a multi-modal AI pipeline. For visual verification, I integrated YOLOv11 via Roboflow to validate genuine grievances like potholes or sewage issues, filtering out low-quality noise. For textual analysis, I utilized the mDeBERTa-v3-base-mnli-xnli model from HuggingFace to classify complaints by department and severity. One of the most challenging aspects was creating a Priority Scoring Engine that computes a 1.0–10.0 severity score by combining image confidence with NLP urgency labels. Additionally, I implemented DBSCAN (Density-Based Spatial Clustering of Applications with Noise) to group complaints into geospatial hotspots, providing administrators with a heatmap of recurring issues.

Beyond the ML models, I prioritized engineering maturity by implementing AES-256-GCM encryption for PII security and a FastAPI-based backend. This project taught me that ML doesn't exist in a vacuum; it requires a robust data pipeline, secure storage, and a deep understanding of how different models (CV, NLP, and Clustering) can work in tandem to create a decision-support system.

Despite these achievements, I recognize significant gaps in my knowledge. While I can leverage pre-trained models like mDeBERTa, I want to move from being a "consumer" of these models to understanding the underlying mathematical optimizations and high-dimensional geometry that drive them. I am particularly interested in the transition from traditional NLP to the scaling laws of Large Language Models (LLMs). Furthermore, I seek to learn about ML at Scale—specifically how Amazon manages model drift and ensures low-latency inference when processing millions of data points across diverse geographies.

The Amazon ML Summer School is the perfect bridge for my goals. Amazon's focus on the science behind the technology aligns with my desire to master the theoretical foundations of Deep Learning. I am particularly excited about the sessions on Probabilistic Graphical Models and LLMs, which are critical for the next evolution of Nivaran—predictive surge forecasting.

My experience building a full-stack, AI-driven civic tool shows that I have the discipline and technical foundation to excel. I am not just a coder; I am a builder who values accuracy, security, and scalability. I am eager to contribute my perspective to the AMLSS 2026 cohort and learn from the world-class scientists at Amazon.